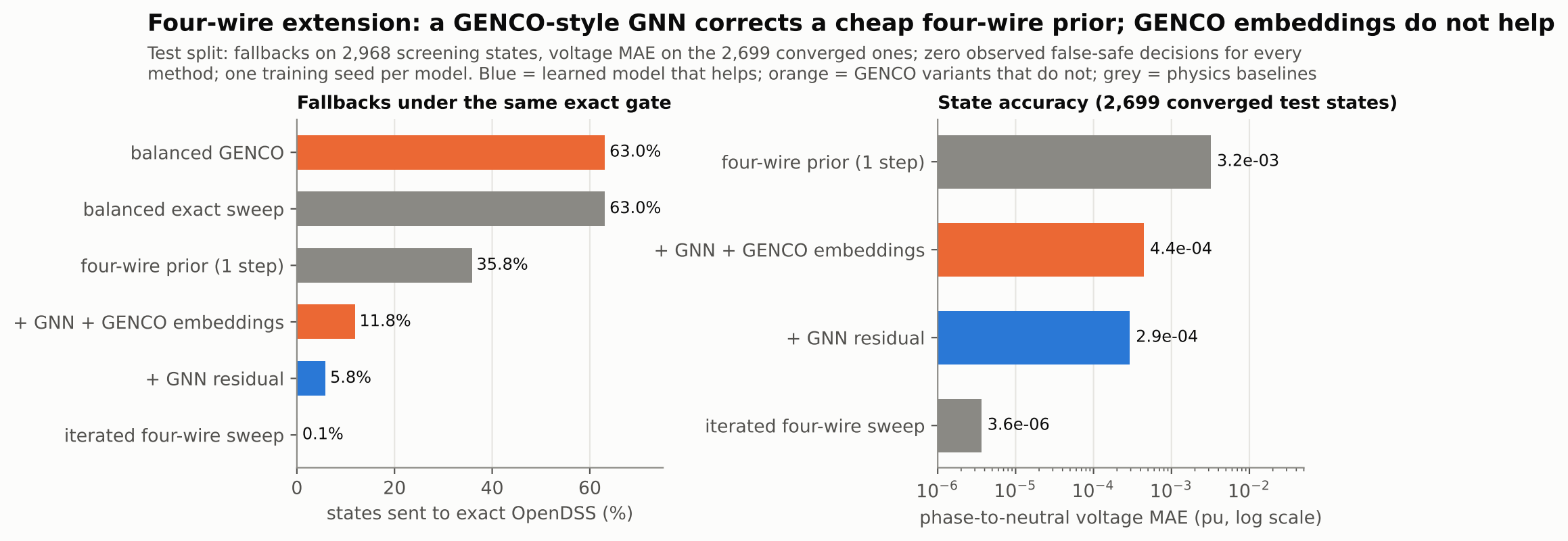
Four-wire extension: a GENCO-style GNN corrects a cheap four-wire prior; GENCO embeddings do not help
Test split: fallbacks on 2,968 screening states, voltage MAE on the 2,699 converged ones; zero observed false-safe decisions for every method; one training seed per model. Blue = learned model that helps; orange = GENCO variants that do not; grey = physics baselines
Fallbacks under the same exact gate
balanced GENCO
63.0%
balanced exact sweep
63.0%
four-wire prior (1 step)
35.8%
+ GNN + GENCO embeddings
11.8%
+ GNN residual
5.8%
iterated four-wire sweep
0.1%
0
20
40
60
states sent to exact OpenDSS (%)
State accuracy (2,699 converged test states)
four-wire prior (1 step)
3.2e-03
+ GNN + GENCO embeddings
4.4e-04
+ GNN residual
2.9e-04
iterated four-wire sweep
3.6e-06
$10^{-6}$
$10^{-5}$
$10^{-4}$
$10^{-3}$
$10^{-2}$
phase-to-neutral voltage MAE (pu, log scale)